THE SPECTRUM OF GLOBAL FUNCTIONS OF AN INTEGRAL COVER OF A QUASI-AFFINE SCHEME

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 13973-14010, immediately following the statement at 13960-13971.

Editorial note. Five cross-chapter pointers (Morphisms, Properties) are recorded in assembly.external_references.

Editorial note. The cartesian square of the statement is displayed verbatim in the statement span.

Allowed standard prerequisites (editorial).

- Integral morphisms are universally closed. Morphisms, lemma morphisms-lemma-integral-universally-closed, cited in the proof. Provides closedness of the morphism, which is what makes the scheme-theoretic image argument work.
- Scheme-theoretic image of a quasi-compact morphism. Morphisms, lemma morphisms-lemma-quasi-compact-scheme-theoretic-image, cited at proof lines 14007-14008. Identifies the image with the spectrum of global functions, giving density of U in it.
- Openness of the structural arrows for quasi-affine schemes. Properties, lemma properties-lemma-quasi-affine, cited at proof lines 13997-13998. Shows the vertical arrows of the square are open, so that the comparison morphism j is an open immersion.
- Composition of quasi-affine morphisms. Morphisms, lemma morphisms-lemma-composition-quasi-affine, cited in the proof. Gives quasi-affineness of U once the cartesian square is established.

Source reference. Original reference 01SN: focused-seeds/geometry/S9-morph-integral-over-quasi-affine/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01W6: focused-seeds/geometry/S9-morph-integral-over-quasi-affine/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01WM: focused-seeds/geometry/S9-morph-integral-over-quasi-affine/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01R8: focused-seeds/geometry/S9-morph-integral-over-quasi-affine/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01P9: focused-seeds/geometry/S9-morph-integral-over-quasi-affine/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let X be a quasi-affine scheme. Let $f : U \to X$ be an integral morphism. Then U is quasi-affine and the diagram*

$$\begin{array}{ccc} U & \longrightarrow & \mathrm{Spec}(\Gamma(U, \mathcal{O}_U)) \\ \downarrow & & \downarrow \\ X & \longrightarrow & \mathrm{Spec}(\Gamma(X, \mathcal{O}_X)) \end{array}$$

is cartesian.

Proof. The scheme U is quasi-affine because integral morphisms are affine, affine morphisms are quasi-affine, a scheme is quasi-affine if and only if the structure morphism to $\mathrm{Spec}(\mathbf{Z})$ is quasi-affine, and compositions of quasi-affine morphisms are quasi-affine. The first two statements follow immediately from the definition and the third is Morphisms, Lemma 01SN. Set $U' = X \times_{\mathrm{Spec}(\Gamma(X, \mathcal{O}_X))} \mathrm{Spec}(\Gamma(U, \mathcal{O}_U))$ and consider the extended diagram

$$\begin{array}{ccccc} U & \xrightarrow{j} & U' & \longrightarrow & \mathrm{Spec}(\Gamma(U, \mathcal{O}_U)) \\ & \searrow & \downarrow & & \downarrow \\ & & X & \longrightarrow & \mathrm{Spec}(\Gamma(X, \mathcal{O}_X)) \end{array}$$

The morphism j is closed by Morphisms, Lemma 01W6 combined with the fact that an integral morphism is universally closed (Morphisms, Lemma 01WM) and the fact that the vertical arrows are in the diagram are separated. On the other hand, j is open because the horizontal arrows in the diagram of the lemma are open by Properties, Lemma 01P9. Thus j identifies U with an open and closed subscheme of U'. If $U \neq U'$ then U isn't dense in U' and a fortiori not dense in the spectrum of $\Gamma(U, \mathcal{O}_U)$. However, the scheme theoretic image of U in $\mathrm{Spec}(\Gamma(U, \mathcal{O}_U))$ is $\mathrm{Spec}(\Gamma(U, \mathcal{O}_U))$ because any ideal in $\Gamma(U, \mathcal{O}_U)$ cutting out a closed subscheme through which U factors would have to be zero. Hence U is dense in $\mathrm{Spec}(\Gamma(U, \mathcal{O}_U))$ for example by Morphisms, Lemma 01R8. Thus $U = U'$ and we win. □